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SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2025

COMMON TO B.E – CSE / B.TECH – IT & M.TECH - CSE

20MA203 - LINEAR ALGEBRA AND DIFFERENTIAL CALCULUS

Answer ALL questions

Duration: 3 Hours

Maximum: 100 Marks

PART - A

(7x 1 = 7 Marks)

1. **CO1** Identify the sum and product of Eigenvalues of the matrix $\begin{pmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{pmatrix}$
a) (10, 20) b) (10, 30) c) (30, 10) d) (10, 25)
2. **CO2** The set of vectors $\{(1, 0), (0, 1)\}$ is _____.
a) linearly dependent b) linearly independent but not span R^2
c) linearly dependent and span R^2 d) basis of R^2
3. **CO3** A function $T: V \rightarrow V$ is called _____ transformation if $T(x) = x$, for all $x \in V$
a) Zero b) rotation c) projection d) Identity
4. **CO2** The dimension of the null space $N(A)$ is called the _____ of A
a) rank b) nullity c) row space d) column space
5. **CO4** Select the parametric equations of the parabola $x^2 = 4ay$
a) $x = a \sec \theta, y = b \tan \theta$ b) $x = a \cos \theta, y = b \sin \theta$
c) $x = 2at, y = at^2$ d) $x = at^2, y = 2at$
6. **CO4** The curvature of the straight line is _____.
a) 0 b) -1 c) 1 d) ∞

7. **CO5** If $u = x^y$ then $\frac{\partial u}{\partial y}$ is _____.

a) yx^{y-1}

b) xy^{x-1}

c) $y^x \log y$

d) $x^y \log x$

Fill in the blanks:

(3x 1 = 3 Marks)

8. **CO1** The nature of the quadratic form $y_1^2 + 2y_2^2 - 3y_3^2$ is _____.

9. **CO2** If the vectors (1,-2) and (-2,x) are linearly dependent then the value of x is _____.

10. **CO4** The radius of curvature of the circle $x^2 + y^2 - 36 = 0$ is _____.

PART - B

(5x 3 = 15 Marks)

11. **CO1** If 3 and 15 are the eigenvalues of $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$, find $|A|$ without expanding the determinant.

12. **CO2** Examine the linear dependence or independence of the following vectors

$$v_1 = (1, 2, 3), v_2 = (2, 5, 7), v_3 = (1, 3, 5)$$

13. **CO3** Show that the mapping $T : R^3 \rightarrow R^3$ defined by $T(x, y, z) = (x, y, 0)$ is Linear.

14. **CO4** Find the radius of curvature at $x = \frac{\pi}{2}$ on the curve $y = 4 \sin x$

15. **CO5** If $x = u(1+v)$ and $y = v(1+u)$, find $\frac{\partial(x,y)}{\partial(u,v)}$

PART - C

(5 x 15 = 75 Marks)

Q.No: 16 Compulsory Question

Q.No: 17 to 22 Answer any FOUR Questions

Compulsory Question:

16. **CO1** i) Make use of orthogonal transformation to reduce the quadratic form (15)
 $2x_1^2 + x_2^2 + x_3^2 + 2x_1x_2 - 2x_1x_3 - 4x_2x_3$ to canonical form. Also determine the rank, index, signature and nature of quadratic form.

17. **CO2** i) Prove that the vectors $v_1 = (2, 9, 0)$, $v_2 = (1, 2, 1)$ and $v_3 = (3, 3, 4)$ form a (7)
 basis for R^3

- CO2** ii) Determine bases for the row space, the column space and the null space of the matrix A , where (8)

$$A = \begin{pmatrix} 2 & 4 & -3 & -6 \\ 7 & 14 & -6 & -3 \\ -2 & -4 & 1 & -2 \\ 2 & 4 & -2 & -2 \end{pmatrix}.$$

Also find the rank and nullity of the matrix

18. **CO3** i) Let $T : P_2(R) \rightarrow P_2(R)$ be the linear transformation defined by (7)

$T(f) = (3+x)f' + 2f$ and $S : P_2(R) \rightarrow R^3$ defined by $S(a+bx+cx^2) = (a-b, a+b, c)$. For a basis $\alpha = \{1, x, x^2\}$ for $P_2(R)$ and the standard basis $\beta = \{e_1, e_2, e_3\}$ for R^3 , compute $[S]_{\alpha}^{\beta}$ and $[T]_{\alpha}$.

- CO3** ii) Consider the following two bases of R^3 . (8)

$\alpha = \{e_1, e_2, e_3\} = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ and

$\beta = \{u_1, u_2, u_3\} = \{(1, 0, 1), (2, 1, 2), (1, 2, 2)\}$

Find the change of basis matrix from the basis α to the basis β .

19. **CO3** i) Let $w_1 = (1,0), w_2 = (2,-1), w_3 = (4,3)$ be three vectors in R^2 . Let (7)

$\alpha = \{e_1, e_2, e_3\}$ be the standard basis for the 3-space R^3 and let $T : R^3 \rightarrow R^2$ be the Linear Transformation defined by $T(e_1) = w_1, T(e_2) = w_2, T(e_3) = w_3$. Find a formula for $T(x_1, x_2, x_3)$ and then use it to compute $T(2, -3, 5)$.

- CO4** ii) Find the equation of the circle of curvature of the curve $xy = c^2$ at (c, c) (8)

20. **CO4** Find the evolute of the parabola $y^2 = 4ax$ (15)

21. **CO1** i) Identify the eigenvalues and eigenvectors of the matrix (8)

$$A = \begin{pmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{pmatrix}$$

- CO5** ii) A rectangular box open at the top is to have a volume of 32 cc. Use Lagrange's multiplier method to find the dimensions of the box that requires the least material for its construction. (7)

22. **CO5** i) Use Taylor's series to expand the function $e^x \cos y$ about the point (8)

$$\left(0, \frac{\pi}{2}\right) \text{ up to third degree terms.}$$

- CO5** ii) Analyze the function $f(x, y) = x^3 + 3xy^2 - 15x^2 - 15y^2 + 72x$ to obtain the stationary points and hence find the maxima and minima. (7)
